

SHUBHANKAR SAHAI

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EDUCATION

University of California, San Diego

PhD Student, Mathematics

September 2021-June 2026 (expected)

Advisor: Kiran Kedlaya

University of Michigan, Ann Arbor

Master of Science, Mathematics

January 2020 - December 2020

University of Michigan, Ann Arbor

Bachelor of Science, Honors Mathematics and Interdisciplinary Physics

September 2014 - December 2017

High Honors

OTHER POSITIONS

Institute for Advanced Study, Princeton

Visiting Graduate Student, School of Mathematics

September 2023-December 2023 and March 2024-April 2024

Sponsor: Kiran Kedlaya

RESEARCH INTERESTS

Arithmetic and algebraic geometry, commutative algebra, homotopy theory

RECOGNITION

James B. Ax Fellowship, UCSD

September, 2021

PAPERS/PREPRINTS

Duality of differential operators and algebraic de Rham cohomology

2024

with Caleb Ji, Casimir Kothari, Oliver Li, Sveta Makarova and Sridhar Venkatesh

Proves a Serre duality result for differential operators. Applied to terms of the E_1 -Hodge-de Rham spectral sequence for smooth proper morphisms, gives another proof of relative Poincaré duality for relative de Rham cohomology.

Preprint available at <https://arxiv.org/abs/2409.14322>

Derived algebras on formal stacks and prismatic gauges.

2026

Available <https://arxiv.org/abs/2602.08942>.

Develops some derived formal geometry to study the theory of derived algebras in quasi-coherent sheaves on derived formal stacks. Ends with Simpson style classification results for derived algebras on stacks modeling the Nygaard filtered prismaticization of perfectoid rings. This has applications to the paper ‘The relative Nygaard filtered prismaticization’.

De Rham affineness of the Nygaard filtered prismaticization in positive characteristic.

2025

Submitted. Preprint available at <https://arxiv.org/abs/2512.19348>.

Names a stronger affineness property (than Toën’s affineness) exhibited by the Nygaard filtered prismaticization in positive characteristic as *de Rham affineness*. Shows that the Nygaard filtered prismaticization in positive characteristic satisfies it.

Relative Poincaré duality for differential operators.

2025

Draft available on request.

with Caleb Ji

Extends results of our previous paper for arbitrary bounded length differential operators along proper smooth morphisms of qcqs schemes. Further proves a duality result with coefficients in bounded above complexes on the base. This answers a question due to Grothendieck in his prenotes to Hartshorne's Residues and Duality. We further construct stable ∞ -categories of differential operators using homotopical techniques.

The relative Nygaard filtered prismaticization.

2026

Draft available on request.

Studies the Nygaard filtered prismaticization relative to a bounded prism. Among other things, shows that this stack is determined by its cohomology ring in a precise sense using the theory developed in 'Derived algebras in prismatic gauges'. This has applications to the paper 'The syntomification of F -liftable schemes'.

The syntomification of F -liftable schemes.

2026

In preparation.

Using the theory developed in 'The relative Nygaard filtered prismaticization' we show that the Nygaard filtered prismaticization of a Frobenius liftable lci scheme is pulled back in a precise sense from the Hodge filtered de Rham stack of its lift. As a consequence we can study the syntomification of such an F -liftable schemes in terms of Fontaine-Laffaille modules in the sense of Faltings. Some open questions about constructing isomorphisms of ring stacks parametrising certain p -adic cohomology theories are discussed.

UNDERGRAD RESEARCH

Composition tableaux basis for Schur functors and the Plücker algebra

2020

Preprint available at <https://arxiv.org/abs/1811.10687>

CONFERENCES ATTENDED/ WILL ATTEND

2025 Summer Research Institute in Algebraic Geometry

July 14-August 1, 2025

Colorado State University in Fort Collins, CO, USA

Arithmetic, K -theory and algebraic cycles

May 27-31, 2025

Ohio State University, Ohio.

Informal seminar on p -adic geometry

September 16-20, 2024

University of Chicago, Chicago.

Local Systems in Algebraic Geometry

May 7-10, 2024

Ohio State University, Ohio.

Special Year on p -adic Arithmetic Geometry, Spring Workshop

11-15 March 2024

IAS, Princeton

Special Year on p -adic Arithmetic Geometry, Fall Workshop

13-17 November 2023

IAS, Princeton

Stacks Project Workshop 2023

7-11 August 2023

Ann Arbor, Michigan

Mentor: Aise Johan de Jong

IHÉS summer school on algebraic K -theory

9-21 July 2023

IHÉS, Bures-sur-Yvette

Algebraic geometry and cohomology in mixed characteristic

15-19 May 2023

Northwestern University, Evanston

TEACHING

At UCSD

TA for Math 20-D (Differential equations)

Fall 2022

TA for Math 20-E (Vector calculus)	<i>Fall 2021, Winter 2022, Spring 2022</i>
TA for Math 109 (Introduction to mathematical thinking)	<i>Winter 2025</i>
TA for Math 154 (Discrete mathematics and graph theory)	<i>Winter 2026</i>
TA for Math 187-A (Cryptography)	<i>Winter 2023, Spring 2023, Winter 2024, Spring 2024, Fall 2024, Spring 2025, Fall 2025</i>

SERVICE

At UCSD

Math Graduate Student Council (MGSC) mentor Mentoring an early grad student at UCSD.	<i>2025-2026</i>
Math Graduate Student Council (MGSC) mentorship chair Ran the mentorship program for the Graduate Student Council. This included pairing first year grad students with senior grad students and other relevant organizational duties.	<i>2024-2025</i>
RTG reading course in Local Class Field Theory Students mentored- Jonathan Cummins, Xun Gong, Fuxiang Yang.	<i>Spring 2023</i>
RTG reading course in commutative and homomological algebra. Students mentored- Yu Sheng. A somewhat intense reading course in commutative and homological algebra culminating in Huneke-Lyubeznik's proof of existence of big Cohen-Macaulay algebras.	<i>Winter-Spring 2025</i>
Founder and organiser for Student Arithmetic Geometry seminar and Discord server The aim is to provide some structure and help to graduate students in arithmetic geometry at UCSD. Among other things, provides a structures space for beginning students to organise learning seminars, etc.	<i>Summer 2022-Present</i>

At Michigan

Minicourse on Algebraic K-theory Summer graduate student mini-course series at Michigan.	<i>June 24-June 28, 2019</i>
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SEMINARS (CO) ORGANISED

<i>Perverse sheaves (aka BBDG) seminar</i> UCSD/Boston University jointly organised with Matt Broe	<i>Summer 2025</i>
<i>Perverse sheaves in commutative algebra reading group</i> UCSD (I gave all the talks).	<i>Winter and Spring 2025</i>
<i>étale K-theory seminar</i> UCSD	<i>Fall 2024</i>
<i>Ogus-Vologodsky seminar</i> UCSD	<i>Fall 2024</i>
<i>p-adic Hodge theory for Rigid Analytic Varieties</i> UCSD/Online	<i>Fall 2023</i>
<i>Derived algebraic geometry</i> UCSD	<i>Spring 2023</i>
<i>Faltings' theorem learning seminar</i> UCSD jointly organised with Nathan Wenger.	<i>Winter 2023</i>
<i>Weil II learning seminar</i> UCSD	<i>Fall 2022</i>

Abelian varieties learning seminar

UCSD

Summer 2022

Prismatic cohomology reading group

UCSD

Winter 2022

Algebraic stacks seminar

UCSD, jointly organised with Jack J. Garzella.

Winter 2022

Crystalline cohomology reading group

UCSD

Fall 2021

OTHER WORK EXPERIENCE.

Lab assistant

February 2018-December 2018

The McMahon Cosmology Lab, University of Michigan, Ann Arbor¹

This job involved writing technical code for machinery which constructs telescope lenses.

Math tutor

February 2019-May 2019

Math Lab, Department of mathematics, University of Michigan, Ann Arbor.

¹Now at the University of Chicago.